

Complete mixed arithmetic efficiently

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Calculate $3,250 \div 25 + 4.8 \times 6$. Copy or complete the first step.

2. Calculate $\frac{2}{3}$ of 450, then subtract 75.

3. Calculate $6,000 - 48 \times 75$.

4. Miro says: Miro carries a method from one number type into the next without checking place value or equivalence. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate $3,250 \div 25 + 4.8 \times 6$. Copy or complete the first step.
Answer 158.8.
2. Calculate $\frac{2}{3}$ of 450, then subtract 75.
Answer 225.
3. Calculate $6,000 - 48 \times 75$.
Answer 2,400.
4. Miro says: Miro carries a method from one number type into the next without checking place value or equivalence. Is this correct? Explain.
Answer Pause at each operation and identify the number representation before choosing the method.

Complete mixed arithmetic efficiently

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Calculate $3,250 \div 25 + 4.8 \times 6$.

6. Check this result using an inverse, estimate or known fact:
Calculate $6,000 - 48 \times 75$.

2. Calculate $\frac{2}{3}$ of 450, then subtract 75.

7. Write a mixed calculation with answer 100 using a fraction and a whole-number operation.

3. Calculate $6,000 - 48 \times 75$.

8. Identify and correct this misconception: Miro carries a method from one number type into the next without checking place value or equivalence.

4. Solve this independently and show every stage: Calculate $3,250 \div 25 + 4.8 \times 6$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Calculate $\frac{2}{3}$ of 450, then subtract 75.

10. Write and solve a similar question to: Calculate $6,000 - 48 \times 75$.

Teacher answers and checking prompts

1. Calculate $3,250 \div 25 + 4.8 \times 6$.
Answer 158.8.
2. Calculate $\frac{2}{3}$ of 450, then subtract 75.
Answer 225.
3. Calculate $6,000 - 48 \times 75$.
Answer 2,400.
4. Solve this independently and show every stage: Calculate $3,250 \div 25 + 4.8 \times 6$.
Answer 158.8.
5. Use a second method or representation for: Calculate $\frac{2}{3}$ of 450, then subtract 75.
Answer Expected result: 225. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate $6,000 - 48 \times 75$.
Answer Expected result: 2,400. A valid check must be shown.
7. Write a mixed calculation with answer 100 using a fraction and a whole-number operation.
Answer Answers vary, for example $\frac{1}{2}$ of $240 - 20 = 100$.
8. Identify and correct this misconception: Miro carries a method from one number type into the next without checking place value or equivalence.
Answer Pause at each operation and identify the number representation before choosing the method.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate $6,000 - 48 \times 75$.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Complete mixed arithmetic efficiently

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Write a mixed calculation with answer 100 using a fraction and a whole-number operation.

6. Create a new example that tests the same idea as: Calculate $6,000 - 48 \times 75$.

2. Prove the answer to this question, rather than only calculating it: Calculate $3,250 / 25 + 4.8 \times 6$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Calculate $\frac{2}{3}$ of 450, then subtract 75.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 2,400.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro carries a method from one number type into the next without checking place value or equivalence.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Write a mixed calculation with answer 100 using a fraction and a whole-number operation.
Answer Answers vary, for example $\frac{1}{2}$ of $240 - 20 = 100$.
2. Prove the answer to this question, rather than only calculating it: Calculate $3,250 \div 25 + 4.8 \times 6$.
Answer Expected result: 158.8. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Calculate $\frac{2}{3}$ of 450, then subtract 75.
Answer Expected result: 225. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 2,400.
Answer One valid reconstruction is: Calculate $6,000 - 48 \times 75$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro carries a method from one number type into the next without checking place value or equivalence.
Answer Pause at each operation and identify the number representation before choosing the method.
6. Create a new example that tests the same idea as: Calculate $6,000 - 48 \times 75$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.